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Inventor(s)

Harben; Grover

Poultry Shackle

Abstract

An articulating, rotating poultry shackle. The shackle has a central member conventionally coupled to a picking line of the processing facility, a fixed hanging portion rigidly coupled to the central member, and an articulating hanging portion rotatably coupled to the fixed hanging portion. A mounting assembly coupled to the central member rotatably attaches the central member to the picking line such that at least three degrees of rotational freedom are provided to the articulating hanging portion of the shackle.

Inventors: Harben; Grover (Gainesville, GA)

Applicant: Harben; Grover (Gainesville, GA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of and priority to U.S. Provisional Patent 63/559,394, filed Feb. 29, 2024, which is incorporated in its entirety in this document by reference.

FIELD OF THE INVENTION

[0002] This invention relates to devices, systems and methods for hanging live poultry. More particularly, this invention relates to an articulating poultry shackle, a rotating poultry shackle, and an articulating, rotating poultry shackle as well as systems and methods for hanging live poultry on shackles for processing.

BACKGROUND OF THE INVENTION

[0003] In poultry processing plants, birds are typically processed in an inverted position with their hocks suspended by shackles pivotally mounted to an overhead conveyor line. Conventionally, the shackles for hanging the bird are rigid shackles coupled to a rail and or chain picking line that move the shackles through the processing facility. That is, a conventional shackle is a rigid, fixed shackle that is not rotatable. This requires that the bird to be placed into the shackle be positioned in a particular position. However, positioning the bird in the particular position can be difficult and requires restraining the bird, orienting the bird correctly relative to the shackles of the conveyor, and placing the bird into the shackles. Therefore, there is a need in the industry for a shackle that is rotatable and/or articulable to provide more flexibility in the way birds can be placed into the shackle. The present invention fulfills these needs and provides further related advantages as describe herein.

SUMMARY

[0004] The application relates to devices, systems and methods for hanging live poultry. More particularly, this application relates to an articulating poultry shackle, a rotating poultry shackle, and an articulating, rotating poultry shackle as well as systems and methods for hanging live poultry on shackles for processing.

[0005] In one aspect, the articulating shackle comprise a central member conventionally coupled to a picking line of the processing facility, a fixed hanging portion rigidly coupled to the central member, and an articulating hanging portion rotatably coupled to the fixed hanging portion. In another aspect, the rotating shackle can comprise the central member, the fixed hanging portion rigidly coupled to the central member and a mounting assembly coupled to the central member for rotatably attaching the central member to the picking line of the processing facility. In still another aspect, the articulating, rotating shackle can comprise the central member, the fixed hanging portion rigidly coupled to the central member, the articulating hanging portion rotatably coupled to the fixed hanging portion, and the mounting assembly coupled to the central member for rotatably attaching the central member to the picking line.

[0006] The shackle can have one, two, three or more than three rotational degrees of freedom. For example, the rotational degrees of freedom can comprise at least one of: 1) the articulated hanging portion rotating relative to the fixed hanging portion about a pivot axis, 2) the articulated hanging portion and the fixed hanging portion rotating relative to a mounting plate about a longitudinal axis; and 3) the articulated hanging portion and the fixed hanging portion rotating relative to the mounting plate about a pivot axis.

[0007] Additional advantages of the invention will be set forth in part in the description which follows, and in part will be obvious from the description, or may be learned by practice of the invention. The advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed.

[0008] Related methods of operation are also provided. Other apparatuses, methods, systems, features, and advantages of the poultry shackle will be or become apparent to one with skill in the

art upon examination of the following figures and detailed description. It is intended that all such additional apparatuses, methods, systems, features, and advantages be included within this description, be within the scope of the poultry shackle, and be protected by the accompanying claims.

Description

DESCRIPTION OF THE FIGURES

[0009] FIG. 1 is a front elevational view of an articulating, rotating shackle according to one aspect, in which the shackle is in a first shackle position;

[0010] FIG. 2 is a left elevational view of the articulating, rotating shackle of FIG. 1, in which a portion of the shackle has rotated to a second shackle position;

[0011] FIG. 3 is a right elevational view of the articulating, rotating shackle of FIG. 1;

[0012] FIG. 4 right elevational view of the articulating, rotating shackle of FIG. 1, in which a portion of the shackle has rotated to a third shackle position;

[0013] FIG. 5 is a first perspective view of a mounting assembly of the articulating, rotating shackle of FIG. 1, according to one aspect;

[0014] FIG. 6 is a second perspective view of the mounting assembly of the FIG. 5; and

[0015] FIG. 7 is a third perspective view of the mounting assembly of the FIG. 5.

DESCRIPTION OF THE INVENTION

[0016] The present invention can be understood more readily by reference to the following detailed description, examples, and claims, and their previous and following description. Before the present system, devices, and/or methods are disclosed and described, it is to be understood that this invention is not limited to the specific systems, devices, and/or methods disclosed unless otherwise specified, as such can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

[0017] The following description of the invention is provided as an enabling teaching of the invention in its best, currently known aspect. Those skilled in the relevant art will recognize that many changes can be made to the aspects described, while still obtaining the beneficial results of the present invention. It will also be apparent that some of the desired benefits of the present invention can be obtained by selecting some of the features of the present invention without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present invention are possible and can even be desirable in certain circumstances and are a part of the present invention. Thus, the following description is provided as illustrative of the principles of the present invention and not in limitation thereof.

[0018] As used herein, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to an “element” includes aspects having two or more such elements unless the context clearly indicates otherwise.

[0019] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint and independently of the other endpoint.

[0020] As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance may or may not occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

[0021] The shackle **10** is illustrated in FIGS. **1-7**, according to various aspects. In one aspect, the shackle **10** can comprise a central member **12**, a fixed hanging portion **14** rigidly coupled to the

central member **12**, and an articulating hanging portion **16** rotatably coupled to the fixed hanging portion **14**. In another aspect, the shackle **10** can comprise the central member **12**, the fixed hanging portion **14** rigidly coupled to the central member **12** and a mounting assembly **18** coupled to the central member **12** for rotatably attaching the central member **12** to a picking line (not shown) of the processing facility. In still another aspect, the shackle **10** can comprise the central member **12**, the fixed hanging portion **14** rigidly coupled to the central member **12**, the articulating hanging portion **16** rotatably coupled to the fixed hanging portion **14**, and the mounting assembly **18** coupled to the central member **12** for rotatably attaching the central member **12** to a picking line (not shown) of the processing facility so that the picking line can move the shackle **10** throughout the processing facility.

[0022] The central member **12** can be an elongate member having a first end **20** and an opposed second end **22** such that the central member **12** extends along a first longitudinal axis A.sub.1. The fixed hanging portion **14** can comprise an elongate first fixed leg **24** and an elongate second fixed leg **26** that is spaced from the first fixed leg **24** a predetermined distance **28**. Each of the first fixed leg **24** and the second fixed leg **26** have a proximal end **30** fixedly attached to the first end **20** of the central member **12** and a distal end **32** extending away from the proximal end **30**. In one aspect, at least a portion on the first fixed leg **24** and/or a portion of the second fixed leg **26** can be substantially parallel to the first longitudinal axis A.sub.1.

[0023] The articulating hanging portion **16** can comprise a first rotating leg **34** and a second rotating leg **36** that is spaced from the first rotating leg **34** a predetermined distance **38**. In one aspect, at least a portion on the first rotating leg **34** and/or a portion of the second rotating leg **36** can be substantially parallel to the first longitudinal axis A.sub.1, though it is contemplated that no portion of the first rotating leg **34** and/or the second rotating leg **36** need be substantially parallel to the first longitudinal axis A.sub.1. Each of the first rotating leg **34** and the second rotating leg **36** have a proximal end **40** and an opposed distal end **42** extending away from the proximal end **40**. In another aspect, at least a portion of the first rotating leg **34** can taper towards the second rotating leg **36**, and/or at least a portion of the second rotating leg **36** can taper towards the first rotating leg **34**. That is, in this aspect, the distal ends **42** of the first rotating leg **34** and the second rotating leg **36** can be positioned closer together than the proximal ends **40** of the first rotating leg **34** and the second rotating leg **36**.

[0024] The articulating hanging portion **16** can further comprise a connecting member **44**, a first inner member **46** and a second inner member **48**. The connecting member **44** can fixedly couple the distal end **42** of the first rotating leg **34** to the distal end **42** of the second rotating leg **36**. In one aspect, the connecting member **44** can be substantially perpendicular to the first longitudinal axis A.sub.1 though it is contemplated that the connecting member **44** need not be substantially perpendicular to the first longitudinal axis A.sub.1.

[0025] Each of the first inner member **46** and the second inner member **48** have a distal end **50** fixedly attached to the connecting member **44** and a proximal end **52** extending away from the distal end **50**. In one aspect, the proximal end **52** of the first inner member **46** can be fixedly attached to the proximal end **52** of the second inner member **48**, though the proximal end **52** of the first inner member **46** need not be fixedly attached to the proximal end **52** of the second inner member **48**. In another aspect, at least a portion of the first inner member **46** can taper away from the second inner member **48**, and/or at least a portion of the second inner member **48** can taper away from the first inner member **46**. That is, in this aspect, the proximal ends **52** of the first inner member **46** and the second inner member **48** can be positioned closer together than the distal ends **50** of the first inner member **46** and the second inner member **48**.

[0026] In one aspect, the first rotating leg **34**, the connecting member **44**, and the first inner member **46** cooperate to form a first slot **54**, and the second rotating leg **36**, the connecting member **44**, and the second inner member **48** cooperate to form a second slot **56**. Because of the taper of the first rotating leg **34**, the second rotating leg **36**, the first inner member **46**, and/or the second inner

member **48**, at least a portion of the first slot **54** and the second slot **56** can be a tapered slot having a greater slot width adjacent to the proximal ends **40** of the first rotating leg **34** and the second rotating leg **36** than the slot width adjacent to the distal ends **42** of the first rotating leg **34** and the second rotating leg **36**. In still another aspect, at least a portion of the first inner member **46** can be spaced from the first rotating leg **34** a first predetermined distance, and at least a portion of the second inner member **48** can be spaced from the second rotating leg **46** a second predetermined distance so that the first slot **54** and the second slot **56** are sized to receive the legs of the bird. In this aspect, the first predetermined distance can be substantially the same as the second predetermined distance, though the first predetermined distance need not be substantially the same as the second predetermined distance.

[0027] A central cavity **58** can be defined by at least a portion of the first fixed leg **24**, the second fixed leg **26**, the first rotating leg **34**, the second rotating leg **36** and the connecting member **44**. In one aspect, the central cavity **58** can be sized and shaped to allow the legs of the bird to be inserted into the central cavity **58** before one leg of the bird is urged into the first slot **54** and the other leg of the bird is urged into the second slot **56** such that the bird is secured into the shackle **10**.

[0028] As previously discussed, the shackle **10** can be an articulating shackle such that the first fixed leg **24** and the second fixed leg **26** are fixed relative to the central member **12**, while the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46** and the second inner member **48** can rotate relative to the first fixed leg **24** and the second fixed leg **26**. In one aspect, a first hinge or first pivot point **60** can be formed and/or positioned on the distal end **32** of the first fixed leg **24** and/or the second fixed leg **26**. In this aspect, the first hinge **60** can be configured so that the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46** and the second inner member **48** can rotate about a second pivot axis A.sub.2 relative to the first fixed leg **24** and the second fixed leg **26**. The second pivot axis A.sub.2 can be substantially perpendicular to the first longitudinal axis A.sub.1 in which at least a portion of the central member **12** can be positioned.

[0029] As illustrated in FIG. 2, at least a portion of the first fixed leg **24** and the second fixed leg **26** can extend along the first longitudinal axis A.sub.1. The first hinge or pivot point **60** can allow the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46** and the second inner member **48** to articulate about and between the first longitudinal axis A.sub.1 and a third longitudinal axis A.sub.3 that is substantially perpendicular to the first longitudinal axis A.sub.1. That is, the articulating hanging portion **16** of the shackle **10** can articulate about and between a first shackle position, in which the articulating hanging portion **16** extends along the first longitudinal axis A.sub.1 (as shown in FIG. 1), and a second shackle position, in which the articulating hanging portion **16** extends along the third longitudinal axis A.sub.3 (as shown in FIG. 2). Optionally, it is contemplated that the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46** and the second inner member **48** can articulate beyond the third longitudinal axis A.sub.3 and back towards the first longitudinal axis A.sub.1, at a first angle **62**. As can be appreciated, the first pivot point **60** can be configured so that the first angle **62** can be any angle between about 1 degree and 359 degrees.

[0030] The mounting assembly **18** is shown in more detail in FIGS. 5-7. The mounting assembly **18** can comprise a mounting plate **70** and an attachment element **72**. As will be described more fully below, interaction between the mounting plate **70** and the attachment element **72** can urge the central member **12** to rotate such that the central member **12**, the first fixed leg **24** and the second fixed leg **26** move about and between the first shackle position as shown in FIG. 1, and a third shackle position as shown in FIG. 4.

[0031] The mounting plate **70** can be a substantially flat plate configured for secured attachment to a picking line (not shown) which conveys the shackle **10** around the processing facility. For example, a plurality of bolt or screw holes **76** can be defined from a first face **78** through the mounting plate **70** and to a second face **80** so that the mounting plate **70** can be fixedly attached to

a portion of the picking line. Other means of attaching the mounting plate **70** to the picking line, such as welding, are also contemplated. The mounting plate **70** can have a mounting face **75** positioned in a mounting plane P.sub.1.

[0032] The attachment element **72** comprises a first attachment element **82** configured to be securedly attached to the mounting plate **70**, and an opposed second attachment element **84** configured to be securedly attached to the central member **12**. In one aspect, the second attachment element **84** can be rotatably coupled to the first attachment element **82**. For example, a second hinge or second pivot point **86** can be formed and/or positioned on the first attachment element **82**. In this aspect, the second hinge **86** can be configured so that the central member **12** and the components fixedly attached to the central member **12** such as the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46** and the second inner member **48** can rotate about a fourth pivot axis A.sub.4.

[0033] The second hinge or pivot point **86** can allow the central member **12**, the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46** and the second inner member **48** to rotate about and between the first shackle position in which the first longitudinal axis A.sub.1 is substantially parallel to the mounting plane P.sub.1 (as illustrated in FIG. 1), and the third shackle position in which the first longitudinal axis A.sub.1 is at a second angle **88** relative to the mounting plane P.sub.1 (as illustrated in FIG. 4). As can be appreciated, the second hinge **86** can be configured so that the second angle **88** can be any angle between about 0 degrees and 359 degrees.

[0034] In one aspect, the attachment element **72** can further comprises a tubular sleeve **90** fixedly attached to a portion of the second attachment element **84**. In this aspect, the tubular sleeve **90** can have an inner diameter configured so that at least a portion of the second end **22** of the central member **12** can extend through the tubular sleeve **90**. An end tip **92** positioned on the portion of the second end **22** of the central member **12** that extends through the tubular sleeve **90** can have an end tip diameter greater than the inner diameter of the tubular sleeve **90**, so that once the end tip **92** is inserted through the tubular sleeve **90**, the central member **12** cannot be removed from the tubular sleeve **90**. In another aspect, the portion of the second end **22** of the central member **12** that is positioned within the tubular sleeve **90** can have an outer diameter less than the inner diameter of the tubular sleeve **90**, so that the second end **22** of the central member **12** can rotate relative to the tubular sleeve **90** about the first longitudinal axis A.sub.1. That is, the central member **12** and the components fixedly attached to the central member **12** such as the fixed hanging portion **14** and the articulating hanging portion **16** of the shackle **10** can rotate about the first longitudinal axis A.sub.1.

[0035] The mounting assembly **18** can further comprise a guide plate **94** fixedly attached to a portion of the central member **12**, and a guide pin **96** fixedly attached to the mounting plate **70**. An arcuate guide slot **98** can be defined in a portion of the guide plate **92**, the guide pin **96** being configured to interact with the guide slot **98** for urging the central member **12** in a desired direction and/or orientation.

[0036] The guide plate **94** can be a rigid plate fixedly attached to a portion of the second end **22** of the central member **12**. The guide plate **94** can have a substantially planer guide face **100** positioned in a guiding plane P.sub.2. As illustrated in FIG. 2, the guide face **100** can be at a third angle **102** relative to the first longitudinal axis A.sub.1. As can be appreciated, the guide face **100** can be configured so that the third angle **102** can be any angle between about 0 degrees and 359 degrees.

[0037] With reference again to FIG. 5, the arcuate guide slot **98** can be defined in the guide plate **94**, the arcuate guide slot **98** having a predetermined slot length and slot width. In one aspect, the slot width can remain constant along the length of the guide slot **98**. Optionally, the slot width can vary along the length of the guide slot **98**. In one aspect, the guide slot **98** can have slot walls **104** that are substantially perpendicular to the guide face **100**. Optionally, the slot walls **104** can be at an

acute angle relative to the guide face **100** so that the slot walls **104** taper away or towards the guide face **100**.

[0038] The guide pin **96** coupled to the mounting plate **70** can be configured to fit in the guide slot **98**. That is, the guide pin **96** can have a pin width that is less than the slot width. Interaction between the guide pin **96** and the slot wall **104** of the guide slot **98** can cause the central member **12** to rotate in a predetermined direction about the fourth pivot axis A.sub.4, and/or interaction between the guide pin **96** and the slot wall **104** of the guide slot **98** can cause the central member **12** to rotate about the first longitudinal axis A.sub.1.

[0039] In one aspect, the mounting assembly **18** can further comprise a pushing arm **106** securedly coupled to a portion of the tubular sleeve **90**, and a pushing arm aperture **108** defined therethrough the mounting plate **70**, as shown in FIG. **6**. The pushing arm aperture **108** can be defined in the mounting plate **70** such that the pushing arm aperture **108** has a predetermined aperture length and aperture width. In one aspect, the aperture slot width can remain constant along the length of the pushing arm aperture **108**. Optionally, the aperture width can vary along the length of the pushing arm aperture **108**. In one aspect, the pushing arm aperture **108** can have aperture walls **116** that are substantially perpendicular to the pushing arm face **110**. Optionally, the aperture walls **116** can be at an acute angle relative to the pushing arm face **110** so that the aperture walls **116** taper away or towards the pushing arm face **110**.

[0040] The pushing arm **106** can be a rigid plate fixedly attached to a portion of the tubular sleeve **90**. The pushing arm **106** can have a substantially planar pushing arm face **110** positioned in a pushing arm plane P.sub.3, as shown in FIG. **5**. In one aspect, the pushing arm **106** can be fixedly attached to the tubular sleeve **90** such that the pushing arm plane P.sub.3 is substantially perpendicular to the guide face plane P.sub.2. Optionally, the pushing arm plane P.sub.3 can be at an acute angle relative to the guide face plane P.sub.2. In another aspect, the pushing arm **106** can be an elongate arm extending linearly away from the tubular sleeve **90** in a straight direction. Alternatively, however, the pushing arm **106** can have a first arm portion **112** extending along a fifth longitudinal axis A.sub.5 that is substantially parallel to the third longitudinal axis A.sub.3, and a second arm portion **114** extending away from the fifth longitudinal axis A.sub.5 at an acute angle, as shown in FIG. **7**. Optionally, at least a portion of the pushing arm **106** can be arcuate in shape.

[0041] The pushing arm **106** coupled to the tubular sleeve **90** can be configured to slide through the pushing arm aperture **108** defined in the mounting plate **70**. That is, the pushing arm **106** can have an arm width that is less than the aperture width. As will be described more fully below, translational movement of the pushing arm **106** through the pushing arm aperture **108** can cause the tubular member **90** to urge the central member **12** in a predetermined direction.

[0042] The assembled shackle **10** can have one, two, three or more than three rotational degrees of freedom. For example, the rotational degrees of freedom can comprise at least one of: 1) the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46** and the second inner member **48** can rotate relative to the first fixed leg **24** and the second fixed leg **26** about the second pivot axis A.sub.2; 2) the central member **12**, the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46**, the second inner member **48**, the first fixed leg **24**, the second fixed leg **26** and the guide plate **92** can rotate relative to the tubular sleeve **90** about the first longitudinal axis A.sub.1; 3) the central member **12**, the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46**, the second inner member **48**, the first fixed leg **24**, the second fixed leg **26**, the guide plate, **92**, the tubular sleeve **90**, the pushing arm **106** and the second attachment element **84** can rotate relative to the first attachment element **82** about the fourth pivot axis A.sub.4.

[0043] In the first shackle position, as shown in FIG. **1**, the first longitudinal axis A.sub.1 and the second pivot axis A.sub.2 can both be substantially parallel to the mounting plane P.sub.1 of the mounting face **75**. That is, in the first shackle position, gravity can allow the central member **12** to

hang such that the central member, the first fixed leg **24**, and the second fixed leg **26** extend substantially linearly in a plane substantially parallel to the mounting face **75**. In the first shackle position, the second angle **88** formed between the longitudinal axis A.sub.1 and the mounting plane P.sub.1 can be about 0 degrees.

[0044] In the third shackle position, as shown in FIG. **4**, the first longitudinal axis A.sub.1 can be at an angle greater than 0 degrees relative to the mounting plane P.sub.1 of the mounting face **75**, and the second pivot axis A.sub.2 can be rotated about 90 degrees relative to the mounting plane P.sub.1 of the mounting face **75**. That is, in the third shackle position, the central member **12**, the first fixed leg **24**, and the second fixed leg **26** can be rotated away from the mounting face **75** (by the second angle **88**), while simultaneously, the central member **12**, the first fixed leg **24**, and the second fixed leg **26** can be rotated about the first longitudinal axis A.sub.1 such that the second pivot axis A.sub.2 is also rotated 90 degrees relative to its position in the first shackle position. As such, in the third shackle position, the central cavity **58** defined by the first fixed leg **24**, the second fixed leg **26**, the first rotating leg **34**, the second rotating leg **36** and the connecting member **44** has also rotated 90 degrees.

[0045] For example, in the first shackle position, a bird could be positioned in the central cavity **58** by inserting the legs of the bird through the central cavity **58** in a direction moving towards the mounting plate **70**, while in the third shackle position, a bird could be positioned in the central cavity **58** by inserting the legs of the bird through the central cavity **58** in a direction moving sideways relative to the mounting plate **70**,

[0046] In use, as the shackle **10** moves through the processing facility on a conveyor picking line, portions of the processing facility can be sized and shaped to interact with the shackle. For example and without limitation, a bar, a frame and the like can be positioned to engage the pushing arm **106** as the shackle **10** is conveyed from a first location towards a second location. This engagement of the pushing arm **106** with a bar or frame can cause the pushing arm **106** to urge the shackle from first shackle position to the third shackle position. Similarly, the same or a different bar or frame can engage the articulating hanging portion **16** of the shackle **10** as the shackle **10** is conveyed from a first location towards a second location. This engagement of the articulating hanging portion **16** with a bar or frame can cause the shackle to move from the first shackle position to the second shackle position. As can be appreciated, the articulating hanging portion **16** of the shackle **10** can be move about and between the first shackle position and the second shackle position when the pushing arm **106** is in the first shackle position and when the pushing arm **106** is in the third shackle position. That is, the articulating hanging portion **16** can be rotated about the second pivot axis A.sub.2 regardless of the rotational status of any other components of the shackle **10**.

[0047] As can also be appreciated, in the processing facility, a bird can be positioned so that the feet of the bird are extended. As the shackle **10** approaches the bird, a bar or frame of the processing facility can urge the shackle **10** from the first shackle position to the third shackle position for picking the bird. That is, the shackle **10** can be rotated about 90 degrees so that the feet of the bird can be caught from the side. After catching the bird, the shackle **10** can be conveyed away from the bar or frame so that gravity urges the shackle **10** from the third shackle position back to the first shackle position for conventional conveyance of the bird throughout the facility.

[0048] Further, it is contemplated that the shackle **10** need not have all the components described herein. For example, the shackle **10** can be an articulating shackle only, in which the first hinge or first pivot point **60** can be formed and/or positioned on the distal end **32** of the first fixed leg **24** and/or the second fixed leg **26** so that the first rotating leg **34**, the second rotating leg **36**, the connecting member **44**, the first inner member **46** and the second inner member **48** can rotate about the second pivot axis A.sub.2 but without the rotating mounting assembly **18**. That is, the shackle **10** can be similar to a conventional shackle **10** with the addition of the first hinge **60** to make the shackle **10** an articulating shackle. In another example, the shackle **10** can be a rotating shackle only, in which the central member **12**, the first rotating leg **34**, the second rotating leg **36**, the

connecting member **44**, the first inner member **46**, the second inner member **48**, the first fixed leg **24**, the second fixed leg **26** and the guide plate **92** can rotate relative to the tubular sleeve **90** about the first longitudinal axis A.sub.1 without the first hinge **60**. That is, the shackle **10** can be similar to a conventional shackle with the addition of the mounting assembly **18** to make the shackle a rotating shackle.

[0049] Although several aspects of the invention have been disclosed in the foregoing specification, it is understood by those skilled in the art that many modifications and other aspects of the invention will come to mind to which the invention pertains, having the benefit of the teaching presented in the foregoing description and associated drawings. It is thus understood that the invention is not limited to the specific aspects disclosed hereinabove, and that many modifications and other aspects are intended to be included within the scope of the appended claims. Moreover, although specific terms are employed herein, as well as in the claims that follow, they are used only in a generic and descriptive sense, and not for the purposes of limiting the described invention.

Claims

1. A shackle for hanging live poultry on a picking line of a processing facility, the shackle comprising: a central member coupled to the picking line, the central member being an elongate central member extending along a first longitudinal axis; a fixed hanging portion coupled to the central member; and an articulating hanging portion rotatably coupled to the fixed hanging portion, wherein the articulating hanging portion is rotatable relative to the fixed hanging portion about a second pivot axis that is substantially perpendicular to the first longitudinal axis.
2. The shackle of claim 1, wherein the articulating hanging portion of the shackle is rotatable about the second pivot axis about and between a first shackle position, in which the articulating hanging portion extends along the first longitudinal axis, and a second shackle position, in which the articulating hanging portion **16** extends along the third longitudinal axis.
3. The shackle of claim 2, wherein the central member is rotatably coupled to the picking line by a mounting assembly comprising: a mounting plate fixedly attached to the picking line; and an attachment element comprising a first attachment element fixedly coupled to the central member, and a second attachment element coupled to the central member and rotatably coupled to the first attachment element such that the central member and the fixed hanging portion of the shackle can rotate about a fourth pivot axis that is parallel to the second pivot axis when the articulating hanging portion of the shackle is in the first shackle position.
4. The shackle of claim 3, wherein the mounting plate has a mounting face positioned in a mounting plane, wherein in the first shackle position, the first longitudinal axis is parallel to the mounting plane.
5. The shackle of claim 4, wherein the articulating hanging portion of the shackle is rotatable about the fourth pivot axis and between the first shackle position, and a third shackle position in which the first longitudinal axis is at an acute angle relative to the mounting plane.
6. The shackle of claim 5, wherein the attachment element further comprises a tubular sleeve fixedly coupled to the second attachment element, wherein a second end of the central member extends through the tubular sleeve, and wherein the central member is rotatable relative to the tubular sleeve about the first longitudinal axis.
7. The shackle of claim 6, wherein the articulating hanging portion of the shackle has at least three degrees of rotational freedom.
8. The shackle of claim 7, wherein a first degree of rotational freedom of the articulating hanging portion is rotation of the articulating hanging portion relative to the fixed hanging portion about the second pivot axis, wherein a second degree of rotational freedom of the articulating hanging portion is rotation of the articulating hanging portion relative to the mounting plane about the fourth pivot axis, and wherein a third degree of rotational freedom of the articulating hanging

portion is rotation of the articulating hanging portion relative to the tubular sleeve about the first longitudinal axis.

9. The shackle of claim 8, wherein the mounting assembly further comprises: a guide plate fixedly attached to a portion of the central member, wherein an arcuate guide slot is defined in the guide plate; and a guide pin fixedly attached to the mounting plate, wherein the guide pin interacts with the guide slot to urge the central member in a desired direction.

10. A method for hanging live poultry on a picking line of a processing facility, the method comprising: providing a shackle, the shackle comprising: a central member coupled to the picking line, the central member being an elongate central member extending along a first longitudinal axis; a fixed hanging portion coupled to the central member; and an articulating hanging portion rotatably coupled to the fixed hanging portion, wherein the articulating hanging portion is rotatable relative to the fixed hanging portion about a second pivot axis that is substantially perpendicular to the first longitudinal axis; and inserting legs of the poultry into slots defined in the articulating hanging portion of the shackle.

11. The method of claim 10, wherein the articulating hanging portion of the shackle is rotatable about the second pivot axis about and between a first shackle position, in which the articulating hanging portion extends along the first longitudinal axis, and a second shackle position, in which the articulating hanging portion **16** extends along the third longitudinal axis.

12. The method of claim 11, wherein the central member is rotatably coupled to the picking line by a mounting assembly comprising: a mounting plate fixedly attached to the picking line; and an attachment element comprising a first attachment element fixedly coupled to the central member, and a second attachment element coupled to the central member and rotatably coupled to the first attachment element such that the central member and the fixed hanging portion of the shackle can rotate about a fourth pivot axis that is parallel to the second pivot axis when the articulating hanging portion of the shackle is in the first shackle position.

13. The method of claim 12, wherein the mounting plate has a mounting face positioned in a mounting plane, wherein in the first shackle position, the first longitudinal axis is parallel to the mounting plane.

14. The method of claim 13, wherein the articulating hanging portion of the shackle is rotatable about the fourth pivot axis and between the first shackle position, and a third shackle position in which the first longitudinal axis is at an acute angle relative to the mounting plane.

15. The method of claim 14, wherein the attachment element further comprises a tubular sleeve fixedly coupled to the second attachment element, wherein a second end of the central member extends through the tubular sleeve, and wherein the central member is rotatable relative to the tubular sleeve about the first longitudinal axis.

16. The method of claim 15, wherein the articulating hanging portion of the shackle has at least three degrees of rotational freedom.

17. The method of claim 16, wherein a first degree of rotational freedom of the articulating hanging portion is rotation of the articulating hanging portion relative to the fixed hanging portion about the second pivot axis, wherein a second degree of rotational freedom of the articulating hanging portion is rotation of the articulating hanging portion relative to the mounting plane about the fourth pivot axis, and wherein a third degree of rotational freedom of the articulating hanging portion is rotation of the articulating hanging portion relative to the tubular sleeve about the first longitudinal axis.

18. The method of claim 17, wherein the mounting assembly further comprises: a guide plate fixedly attached to a portion of the central member, wherein an arcuate guide slot is defined in the guide plate; and a guide pin fixedly attached to the mounting plate, wherein the guide pin interacts with the guide slot to urge the central member in a desired direction.

19. A system for moving poultry through a processing facility, the system comprising: at least one shackle coupled to a picking line, the shackle comprising: a central member coupled to the picking

line, the central member being an elongate central member extending along a first longitudinal axis; a fixed hanging portion coupled to the central member; and an articulating hanging portion rotatably coupled to the fixed hanging portion, wherein the articulating hanging portion is rotatable relative to the fixed hanging portion about a second pivot axis that is substantially perpendicular to the first longitudinal axis; and the picking line that conveys the at least one shackle through the processing facility.

20. The system of claim 19, wherein the articulating hanging portion of the shackle is rotatable about the second pivot axis about and between a first shackle position, in which the articulating hanging portion extends along the first longitudinal axis, and a second shackle position, in which the articulating hanging portion **16** extends along the third longitudinal axis.
